

Notes

UX User experience

UI User interface $\Rightarrow$ visual interface seen by the user

Docker

allows you to package an application with everything it needs
(code, dependencies, configurations)

create image

creates and manages containers

image

The image contains everything:

- The application
- libraries
- configuration

container $\Rightarrow$ image running

FastAPI

It is a Python framework for creating (web APIs) (backend services)

User interface design

principles of complexity

complexity of the system should not necessarily be reflected in the complexity of the interface

- The goal is to keep it simple

humane interface

Four laws

(1. The goal is to keep things simple)

- 1.- A computer should not damage your work.
- 2.- It should not waste your time or require more work than is strictly necessary.
- 3.- An interface is human centered if it responds to human needs and considers human frailties
- 4.- The user should control the pace of interaction

Cognetics and locus of attention

history

- GUIs emerged in the 1960s at Xerox Palo Alto
- P.O.O. also appeared around the 1960s.

HCI constants

humans have:

memory: limited to 7 items in short-term memory

Perceptual System: for perceiving the environment.

cognitive System: for processing / thinking

motor System: for executing responses.

cognitive limitations

- Humans have only one conscious stream of thought

Key design principles

Avoid long lists of options / group them together

Avoid modes

User terms and concepts familiar to the user

Determining Content

- Capture / refine actors and users goals.
- Analyze objects and relationships
- Define / refine requirements
- Explore real scenarios
- Tools: use cases, requirements definition, class diagram

Interaction Diagrams, State Diagrams.

interface languages

Recognizable styles with their own vocabulary of objects, actions and visuals:

- form-based applications
- Media players, Text editors
- Intographics, Graphics editors.
- Immersive games
- Web pages, Browsers
- Calendars
- E-commerce sites.

Pattern summary

resumen del patrón

extras on demand

What hide advanced features/options until user requests them

When Complex applications with both novice and expert users; when screen space is limited

Why Keeps interface simple for beginners providing power features for experts; reduces cognitive load.

One-Window drill Down

What Navigate through hierarchical content in a single window, replacing previous content with deeper levels.

When Mobile devices or limited screen space; clear hierarchical information structure

Why Conserves screen space; provides clear navigation path; works well on small screens.

Intriguing Branches

What offer enticing links/paths that invite exploration (visual hints, previews)

When Content-rich sites; when user engagement and discovery are important; e-commerce

Why Encourages exploration; keeps users engaged; helps discover content they didn't know they wanted.

User research

- Different classes of users
- User goals in using the software
- Their skill level using similar software
- Their attitudes toward what you're designing
- Their experiences with previous systems.

user research methods

- direct observation
- interviews
- case studies
- Survey
- Pensong

User (Behavior) Patterns

- Safe exploration
- Instant Gratification
- Satisficing
- changes in midstream
- Deferred choice
- Incremental construction
- habituation
- Microbreaks
- Spatial memory
- Prospective memory
- streamlined repetition
- Keyboard only
- other people's advice & Personal recommendations.